



Machine Learning Practice Repository



Repository Overview

This repository contains my Machine Learning learning journey, hands-on practice notebooks, data analysis exercises, and mini-projects implemented using Python.

The purpose of this repository is to build a strong foundation in Data Science and Machine Learning through practical implementation of concepts such as data cleaning, exploratory data analysis (EDA), data visualization, feature engineering, and machine learning workflows.

As I continue learning, new notebooks, projects, and experiments will be added regularly.



Topics Covered

- Python for Data Science
 - NumPy
 - Pandas
 - Data Cleaning
 - Data Quality Analysis
 - Exploratory Data Analysis (EDA)
 - Data Visualization
 - Statistical Analysis
 - Feature Engineering
 - Machine Learning Fundamentals
 - Model Evaluation
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Technologies Used

- Python 3
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Git & GitHub
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Repository Structure

```
ML-Practice/  
|  
├─ EDA.ipynb  
├─ data_check_graph.ipynb  
├─ Loan payments data.csv  
├─ README.md  
└─ Future Notebooks & Projects
```

Current Notebooks

1. EDA.ipynb

This notebook focuses on Exploratory Data Analysis techniques including:

- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis
- Distribution Analysis
- Correlation Analysis
- Heatmaps
- Boxplots
- Scatterplots
- Data Visualization Techniques

2. data_check_graph.ipynb

This notebook focuses on Data Quality Assessment including:

- Dataset Inspection
- Missing Value Analysis
- Null Count Calculation
- Missing Value Percentage Analysis
- Data Quality Visualization
- Heatmap-Based Analysis

Dataset Used:

- Loan Payments Dataset
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Learning Outcomes

This repository demonstrates practical understanding of:

- Data Exploration
 - Data Cleaning Techniques
 - Data Visualization
 - Statistical Thinking
 - Problem Solving with Python
 - Machine Learning Preparation Workflow
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Repository Goals

- Build a strong Machine Learning foundation
 - Practice concepts through implementation
 - Maintain a structured learning portfolio
 - Document progress consistently
 - Prepare for real-world Data Science and AI/ML projects
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How to Use

1. Clone the repository:

```
git clone https://github.com/<your-username>/ML-Practice.git
```

1. Install required libraries:

```
pip install pandas numpy matplotlib seaborn scikit-learn jupyter
```

1. Launch Jupyter Notebook:

```
jupyter notebook
```

1. Open any notebook and run the cells sequentially.
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Learning Roadmap

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Python
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NumPy
↓
Pandas
↓
Data Cleaning
↓
EDA
↓
Data Visualization
↓
Machine Learning
↓
Deep Learning
```

License

This repository is intended for educational and learning purposes.

Author

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Currently exploring Data Analysis, Machine Learning, Artificial Intelligence, and Data Engineering.

★ This repository will continue to grow as I learn, practice, and build more Machine Learning projects.